Advanced Crux Chemistry

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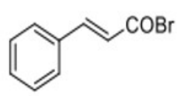
Advanced Crux Chemistry

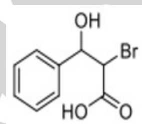
Paragraph for Question 1 and Question 2

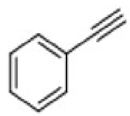
Treatment of benzene with CO/HCl in the presence of anhydrous $AlCl_3$/CuCl followed by reaction with Ac_2O/NaOAc gives compound X as the major product. Compound X upon reaction with Br_2/Na_2CO_3, followed by heating at 473 K with Moist KOH furnishes Y as the major product. Reaction of X with H_2/Pd-C, followed by H_3PO_4 treatment gives Z as the major product.

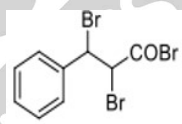
Choose the correct answer.

1. The compound Y is

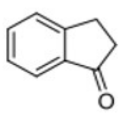
(A) 

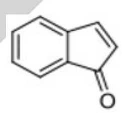
(B) 

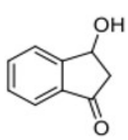
(C) 

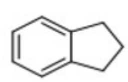
(D) 

2. The compound Z is **[2018(Paragraph)]**

(A) 

(B) 

(C) 

(D) 

3. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to **[2017]**

(A) The physical state of X_2 at atom temperature changes from gas to solid down the group

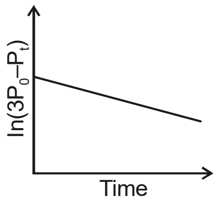
(B) Decrease in HOMO-LUMO gap down the group

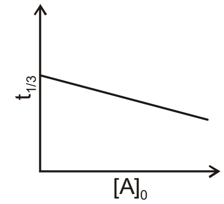
(C) Decrease in π^* - σ^* gap down the group

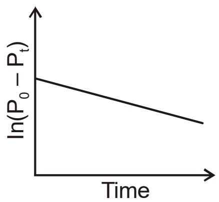
(D) Decrease in ionization energy down the group

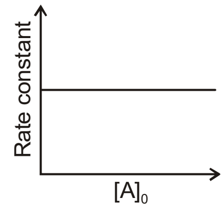
4. For a first order reaction $A(g) \rightarrow 2B(g) + C(g)$ at constant volume and 300 K, the total pressure at the beginning (t = 0) and at time t are P_0 and P_t, respectively. Initially, only A is present with concentration $[A]_0$,

and $t_{1/3}$ is the time required for the partial pressure of A to reach 1/3rd of its initial value. The correct option(s) is(are) (Assume that all these gases behave as ideal gases) **[2018(MCQ)]**

(A) 

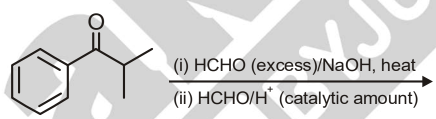
(B) 

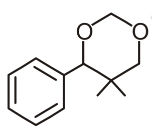
(C) 

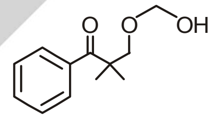
(D) 

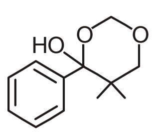
5. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) **[2017(MCQ)]**
   (A) The activation energy of the reaction is unaffected by the value of the steric factor
   (B) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation
   (C) Since P = 4.5, the reaction will not proceed unless an effective catalyst is used
   (D) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

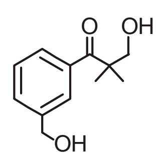
6. The major product of the following reaction sequence is **[2016]**



(A) 

(B) 

(C) 

(D) 

7. An organic compound (P) of molecular formula $C_5H_{10}O$ is treated with dil. H_2SO_4 to give two compounds (Q) and (R) both of which respond iodoform test. The rate of reaction of (P) with dil. H_2SO_4 is 10^{10} faster than the reaction of ethylene with dil. H_2SO_4. Identify the organic compounds, (P), (Q) and (R) and explain the extra reactivity of (P). **[2004]**

8. The correct option(s) regarding the complex $[Co(en)(NH_3)_3(H_2O)]^{3+}$ (en = $H_2NCH_2CH_2NH_2$) is(are)
   (A) It has two geometrical isomers
   (B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
   (C) It is paramagnetic
   (D) It absorbs light at longer wavelength as compared to $[Co(en)(NH_3)_4]^{3+}$

[2018(MCQ)]

9. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
(Atomic numbers: Fe = 26, Ni = 28)
(A) Total number of valence shell electrons at metal centre in $Fe(CO)_5$ or $Ni(CO)_4$ is 16
(B) These are predominantly low spin in nature
(C) Metal-carbon bond strengthens when the oxidation state of the metal is lowered
(D) The carbonyl C-O bond weakens when the oxidation state of the metal is increased

[2018(MCQ)]

10. The electrochemical cell shown below is a concentration cell.
$M|M^{2+}$ (saturated solution of a sparingly soluble salt, MX_2) || M^{2+} (0.001 mol dm^{-3}) |M
The emf of the cell depends on the difference in concentration of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059 V.

(a) The value of ΔG (kJ mol^{-1}) for the given cell is (take 1F = 96500 C mol^{-1})

(A) – 5.7 (B) 5.7
(C) 11.4 (D) –11.4

(b) The solubility product (K_{sp}; $mol^3\ dm^{-9}$) of MX_2 at 298 K based on the information available the given concentration cell is (take 2.303 × R × 298/F = 0.059 V) **[2012(Paragraph)]**

(A) 1×10^{-15} (B) 4×10^{-15}
(C) 1×10^{-12} (D) 4×10^{-12}

11. The standard reduction potential data at 25°C is given below:
$E^o (Fe^{3+}, Fe^{2+}) = +0.77V$; $E^o (Fe^{2+}, Fe) = -0.44V$
$E^o (Cu^{2+}, Cu) = +0.34V$; $E^o (Cu^{+}, Cu) = +0.52V$
$E^o [O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O]$ $E^o [O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-] = +1.23V; = +0.40V$
$E^o (Cr^{3+}, Cr) = -0.74V$; $E^o (Cr^{2+}, Cr) = -0.91V$
Match E^o of the redox pair in List – I with the values given in List – II and select the correct answer using the code given below the lists:

(P) $E^o (Fe^{3+}, Fe)$ (1) –0.18V
(Q) $E^o (4H_2O \rightleftharpoons 4H^+ + 4OH^-)$ (2) –0.4V
(R) $E^o (Cu^{2+} + Cu \rightarrow 2Cu^+)$ (3) –0.04V
(S) $E^o (Cr^{3+}, Cr^{2+})$ (4) –0.83V

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	2	3	4	1
(C)	1	2	3	4
(D)	3	4	1	2

[2013(Matrix)]

12. Dilution processes of different aqueous solutions, with water, are given is LIST – I. The effects of dilution of the solutions on $[H^+]$ are given in LIST – II.

(**Note:** Degree of dissociation (α) of weak acid and weak base is <<1; degree of hydrolysis of salt <<1; $[H^+]$ represents the concentration of H^+ ions)

LIST – I	LIST – II
P. (10 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 60 mL	**1.** The value of $[H^+]$ does not change on dilution
Q. (20 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 80 mL	**2.** The value of $[H^+]$ changes to half of its initial value on dilution
R. (20 mL of 0.1 M HCl + 20 mL of 0.1 M ammonia solution) diluted to 80 mL	**3.** The value of $[H^+]$ changes to two times of its initial value on dilution
S. 10 mL saturated solution of $Ni(OH)_2$ in equilibrium with excess solid $Ni(OH)_2$ is diluted to 20 mL (solid $Ni(OH)_2$ is still present after dilution).	**4.** The value of $[H^+]$ changes to $\frac{1}{\sqrt{2}}$ times of its initial value on dilution
	5. The value of $[H^+]$ changes to $\sqrt{2}$ times of its initial value on dilution

Match each process given in LIST – I with one or more effect(s) in LIST – II. The correct option is

(A) P → 4; Q → 2; R → 3; S → 1

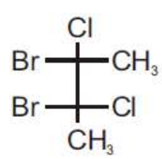
(B) P → 4; Q → 3; R → 2; S → 3

(C) P → 1; Q → 4; R → 5; S → 3

(D) P → 1; Q → 5; R → 4; S → 1

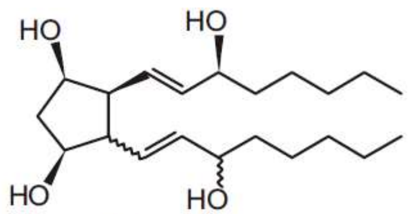
[2018(Matrix)]

13. The Total number(s) of stable conformers with non-zero dipole moment for the following compound is(are)



[2014(Integer)]

14. For the given compound **X,** the total number of optically active stereoisomers is ___ **[2018(Integer)]**

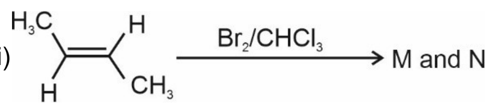


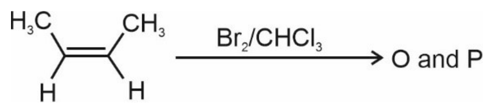
X

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is fixed

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is NOT fixed

15. The correct statement(s) for the following addition reactions is(are) **[2017(MCQ)]**

(i) 

(ii) 

(A) (**M** and **O**) and (**N** and **P**) are two pairs of diastereomers

(B) Bromination proceeds through *trans*-addition in both the reactions

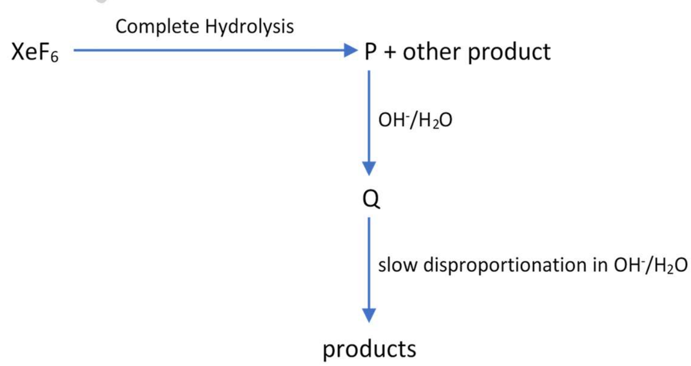
(C) **O** and **P** are identical molecules

(D) (**M** and **O**) and (**N** and **P**) are two pairs of enantiomers

16. An organic compound X, on analysis gives 24.24 percent carbon and 4.04 percent hydrogen. Further, sodium extract of 1.0 g of X gives 2.90 g of silver chloride with acidified silver nitrate solution. The compound X may be represented by two isomeric structures, Y and Z. Y on treatment with aqueous potassium hydroxide solution gives a dihydroxy compound while Z on similar treatment gives ethanal. Find out the molecular formula of X and give the structures of Y and Z. **[1989(Subjective)]**

17. An alkene A ($C_{16}H_{16}$) on ozonolysis gives only one product B (C_8H_8O). Compound B on reaction with NaOH/I_2 yields sodium benzoate. Compound B reacts with KOH/NH_2NH_2 yielding a hydrocarbon C (C_8H_{10}). Write the structure of compounds B and C. Based on this information two isomeric structures can be proposed for alkene A. Write their structures and identify the isomer which on catalytic hydrogenation (H_2/Pd-C) gives a racemic mixture. **[2001(Subjective)]**

18. Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is **[2014(Paragraph)]**



(A) 0 (B) 1

(C) 2 (D) 3

19. All the compounds listed in Column I react with water. Match the result of the respective reactions with the appropriate options listed in Column II.

Column – I		Column – II	
(A)	$(CH_3)_2SiCl_2$	(p)	Hydrogen halide formation
(B)	XeF_4	(q)	Redox reaction
(C)	Cl_2	(r)	Reacts with glass
(D)	VCl_5	(s)	Polymerization
		(t)	O_2 formation

[2010(Matrix)]

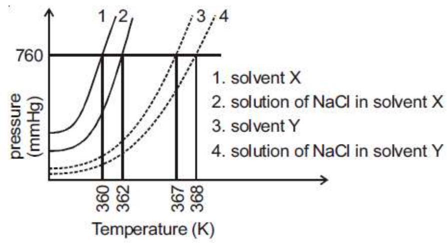
20. Read the following statement and explanation and answer as per the option given below:
Assertion: The first ionization energy of Be is greater than that of B.
Reason: 2p orbital is lower in energy than 2s.
(i) If both assertion and reason are CORRECT, and reason is the CORRECT explanation of the assertion
(ii) If both assertion and reason are CORRECT, but reason is NOT the CORRECT explanation of the assertion
(iii) Assertion is CORRECT but reason is INCORRECT
(iv) If assertion is INCORRECT, but reason is CORRECT **[2000]**

21. To measure the quantity of $MnCl_2$ dissolved in an aqueous solution, it was completely converted to $KMnO_4$ using the reaction, $MnCl_2 + K_2S_2O_8 + H_2O \rightarrow KMnO_4 + H_2SO_4 + HCl$ (equation not balanced). Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeared. The quantity of $MnCl_2$ (in mg) present in the initial solution is ______.
(Atomic weights in g mol^{-1}: Mn = 55, Cl = 35.5) **[2018(Subjective/Integer Type)]**

22. Consider an ionic solid **MX** with NaCl structure. Construct a new structure (**Z**) whose unit cell is constructed from the unit cell of **MX** following the sequential instructions given below. Neglect the charge balance.
(i) Remove all the anions (**X**) except the central one
(ii) Replace all the face centered cations (**M**) by anions (**X**)
(iii) Remove all the corner cations (**M**)
(iv) Replace the central anion (**X**) with cation (**M**)
The value of $\left(\frac{\text{number of anions}}{\text{number of cations}}\right)$ in **Z** is ____. **[2018(Subjective)]**

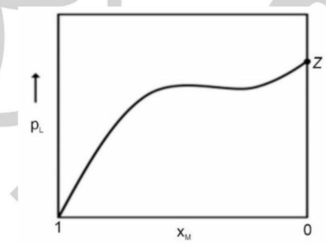
23. 20% of surface sites are occupied by N_2 molecules. The density of surface site is 6.023×10^{14} cm^{-2} and total surface area is 1000 cm^2. The catalyst is heated to 300 K while N_2 completely desorbed into a pressure of 0.001 atm and volume of 2.46 cm^3. Find the number active sites occupied by each N_2 molecule.
[2005(Subjective)]

24. The plot given below shows P-T curves (Where P is the pressure and T is the temperature) for two solvents X and Y and isomolal solutions of NaCl in these of NaCl in these solvents. NaCl completely dissociates in both the solvents



On addition of equal number of moles of a non- volatile solute **S** in equal amount (in kg) of these solvents, the elevation of boiling point of solvent **X** is three times that of solvent **Y**. Solute **S** is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent **Y**, the degree of dimerization in the solvent **X** is __________. **[2018(Subjective)]**

25. For a solution formed by mixing liquids **L** and **M**, the vapour pressure of **L** plotted against the mole fraction of **M** in solution is shown in the following figure. Here x_L and x_M represent mole fractions of **L** and **M**, respectively, in the solution. The correct statement(s) applicable to this system is(are). **[2017(MCQ)]**



(A) Attractive intermolecular interactions between **L-L** in pure liquid **L** and **M-M** in pure liquid **M** are stronger than those between **L-M** when mixed in solution

(B) The point **Z** represents vapour pressure of pure liquid **M** and Raoult's law is obeyed when $x_L \to 0$

(C) The point **Z** represents vapour pressure of pure liquid **L** and Raoult's law is obeyed when $x_L \to 1$

(D) The point **Z** represents vapour pressure of pure liquid **M** and Raoult's law is obeyed when $x_L = 0$ to $x_L = 1$

26. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of the molecular weight of the solute and solvent, $\left(\frac{\mathrm{MW_{solute}}}{\mathrm{MW_{solvent}}}\right)$, is **[2016(Integer)]**

27. The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 2 times. As a result, the diffusion coefficient of this gas increases *x* times. The value of *x* is **[2016(Subjective)]**

28. The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at T = 298 K are

$$\Delta_f G^0[\mathrm{C(graphite)}] = 0\ \mathrm{kJmol^{-1}}$$

$$\Delta_f G^0[\mathrm{C(diamond)}] = 2.9\ \mathrm{kJmol^{-1}}$$

The standard state means that the pressure should be 1 bar, and substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond C[(diamond)] reduces its volume by 2 ×

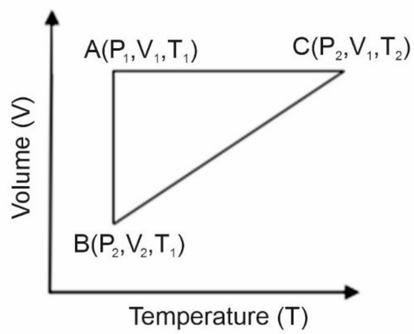
10^{-6} m^3 $mole^{-1}$. If C(graphite) is converted to C(diamond) isothermally at T = 298 K, the pressure at which C (graphite) is in equilibrium with C(diamond), is

[Useful information: 1 J = 1 kg m^2s^{-2}; 1 Pa = 1 kg $m^{-1}s^{-2}$ bar = 10^5 Pa] **[2017(MCQ)]**

(A) 14501 bar (B) 58001 bar

(C) 1450 bar (D) 29001 bar

29. A reversible cyclic process for an ideal gas is shown below. Here, P, V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively **[2018(MCQ)]**



The correct option(s) is(are)

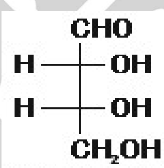
(A) $q_{AC} = \Delta U_{BC}$ and $W_{AB} = P_2(V_2 - V_1)$

(B) $W_{BC} = P_2(V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$

(C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$

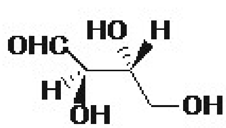
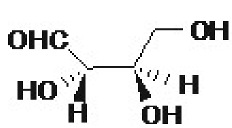
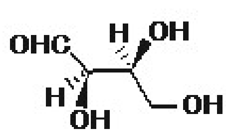
(D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} < \Delta U_{CA}$

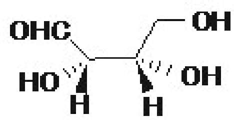
30. The Fischer projection of D-erythrose is shown below.



D-Erythrose

D-Erythrose and its isomers are listed as **P**, **Q**, **R**, and **S** in **Column-I**. Choose the correct relationship of **P**, **Q**, **R**, and **S** with D-erythrose from **Column II**. **[2020(SCQ)]**

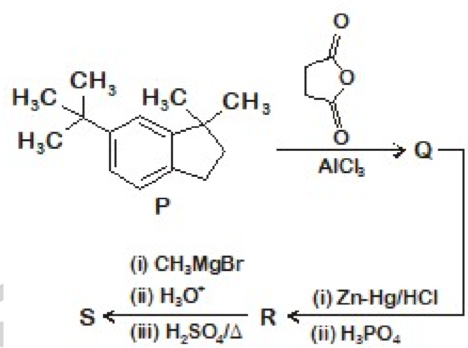
Column-I		**Column-II**
P.		1. Diastereomer
Q.		2. Identical
R.		3. Enantiomer

S. 

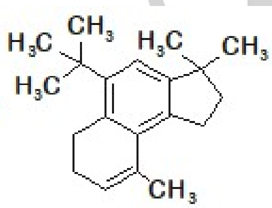
(A) **P → 2, Q → 3, R → 2, S → 2** **(B)** **P → 3, Q → 1, R → 1, S → 2**

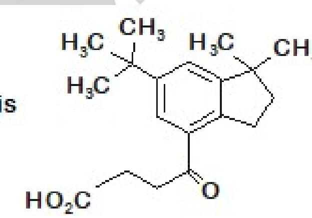
(C) **P → 2, Q → 1, R → 1, S → 3** **(D)** **P → 2, Q → 3, R → 3, S → 1**

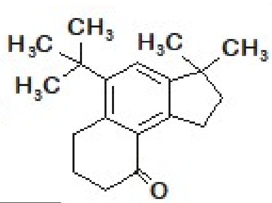
31. **In the reaction scheme shown below, Q, R, and S are the major products.**

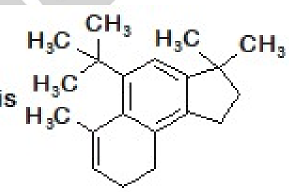


The correct structure of **[2020(MCQ)]**

(A) **S is** 

(B) **Q is** 

(C) **R is** 

(D) **S is** 

32. With respect to hypochlorite, chlorate and perchlorate ions, choose the correct statement(s). **[2020(MCQ)]**

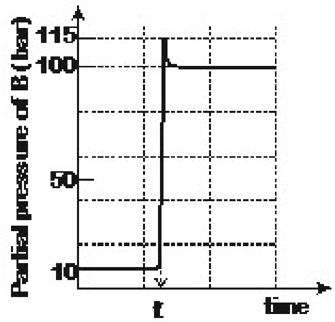
(A) The hypochlorite ion is the strongest conjugate base

(B) The molecular shape of only chlorate ion is influenced by the lone pair of electrons of Cl

(C) The hypochlorite and chlorate ions disproportionate to give rise to identical set of ions

(D) The hypochlorite ion oxidizes the sulfite ion

33. Consider the reaction $A \rightleftharpoons B$ at 1000 K. At time t¢, the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of **A** was maintained at 1 bar. Given below is the plot of the partial pressure of **B** with time. What is the ratio of the standard Gibbs energy of the reaction at 1000 K to that at 2000 K? **[2020(Num.)]**



34. Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$$

The work derived from the cell on the consumption of 1.0 × 10^{-3} mol of $H_2(g)$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulated container. What is the change in the temperature (in K) of the ideal gas?

The standard reduction potentials for the two half-cells are given below.

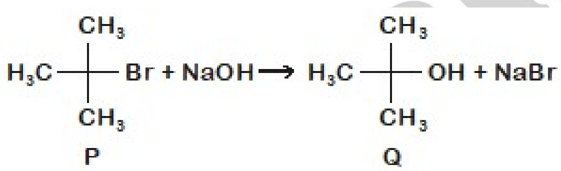
$$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l),\ E^o = 1.23\ V,$$

$$2H^+(aq) + 2e^- \rightarrow H_2(g),\ E^o = 0.00\ V.$$

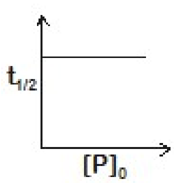
Use F = 96500 C mol^{-1}, R = 8.314 J mol^{-1} K^{-1}. **[2020(Num.)]**

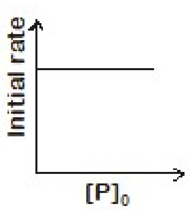
35. Which of the following plots is(are) correct for the given reaction?

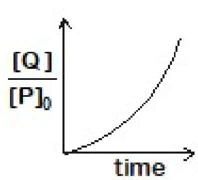
($[P]_0$ is the initial concentration of P)

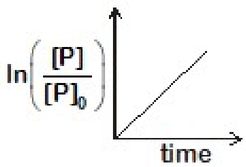


[2020(MCQ)]

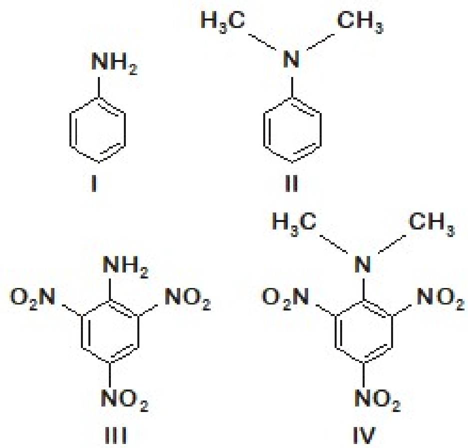
(A) 

(B) 

(C) 

(D) 

36. Consider the following four compounds I, II, III, and IV.



Choose the correct statement(s). **[2020(MCQ)]**

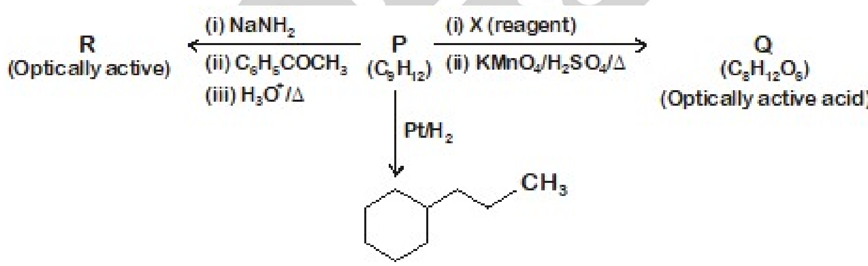
(A) The order of basicity is II > I > III > IV.

(B) The magnitude of pK_b difference between I and II is more than that between III and IV.

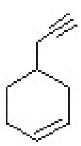
(C) Resonance effect is more in III than in IV.

(D) Steric effect makes compound IV more basic than III.

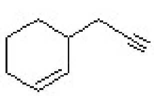
37. Consider the following transformations of a compound P.

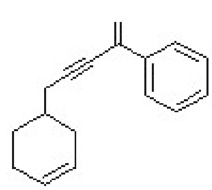


Choose the correct option(s). **[2020(MCQ)]**

(A) P is 

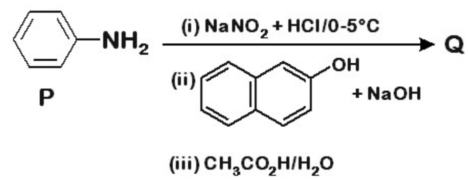
(B) X is Pd-C/quinoline/H_2

(C) P is 

(D) R is 

38. Liquids **A** and **B** form ideal solution for all compositions of **A** and **B** at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of **A** have the total vapor pressures of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid **B** in bar? **[2020(Num.)]**

39. Consider the reaction sequence from **P** to **Q** shown below. The overall yield of the major product **Q** from **P** is 75%. What is the amount in grams of **Q** obtained from 9.3 mL of **P**? (Use density of **P** = 1.00 g mL^{-1}; Molar mass of C = 12.0, H = 1.0, O = 16.0 and N = 14.0 g mol^{-1}) **[2020(Num.)]**



40. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

At 298 K: $\Delta_f H°(SnO_2(s)) = -581.0$ kJ mol^{-1}, $\Delta_f H°(CO_2(g)) = -394.0$ kJ mol^{-1},

$S°(SnO_2(s)) = 56.0$ J K^{-1} mol^{-1}, $S°(Sn(s)) = 52.0$ J K^{-1} mol^{-1},

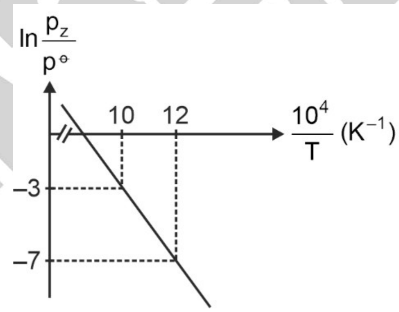
$S°(C(s)) = 6.0$ J K^{-1} mol^{-1}, $S°(CO_2(g)) = 210.0$ J K^{-1} mol^{-1}.

Assume that the enthalpies and the entropies are temperature independent. **[2020(Num.)]**

Question stem for Question Nos. 41 and 42

Question Stem

For the reaction, $X(s) \rightleftharpoons Y(s) + Z(g)$, the plot of $\ln\frac{p_z}{p^{\ominus}}$ versus $\frac{10^4}{T}$ is given below (in solid line), where p_z is the pressure (in bar) of the gas Z at temperature T and $p^{\ominus} = 1$ bar.



(Given, $\frac{d(\ln K)}{d\left(\frac{1}{T}\right)} = -\frac{\Delta H^{\ominus}}{R}$, where the equilibrium constant, $K = \frac{p_z}{p^{\ominus}}$ and the gas constant, R = 8.314 J K^{-1} mol^{-1})

41. The value of standard enthalpy, $\Delta H^{\ominus}$ (in kJ mol^{-1}) for the given reaction is ______. **[2021(Num.)]**

42. The value of $\Delta S^{\ominus}$ (in J K^{-1} mol^{-1}) for the given reaction, at 1000 K is ____. **[2021(Num.)]**

Question stem for Question Nos. 43 and 44

Question Stem

The boiling point of water in a 0.1 molal silver intrate solution (soltuion A) is x°C. To this solution A, an equal volume of 0.1 molal aqueous barium chloride solution is added to make a new solution B. The difference in the boiling points of water in the two solutions A and B is y × 10^{-2} °C.

(Assume: Densities of the solutions A and B are the same as that of water and the soluble salts dissociate completely.

Use: Molal elevation constant (Ebullioscopic constant), K_b = 0.5 K kg mol^{-1}; Boiling point of pure water as 100ºC.)

43. The value of x is ______. **[2021(Num.)]**

44. The value of |y| is _____. **[2021(Num.)]**

45. A mixture of two salts is used to prepare a solution S, which gives the following results:

$$\text{White precipitate(s) only} \xleftarrow[\text{Room temperature}]{\text{Dilute NaOH(aq)}} \underset{\text{(aq solution of the salts)}}{S} \xrightarrow[\text{Room temperature}]{\text{Dilute HCl(aq)}} \text{White precipitate(s) only}$$

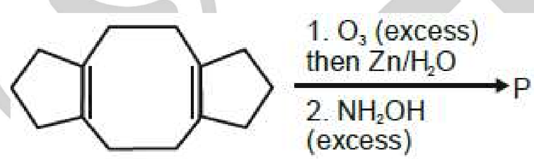
The correct option(s) for the salt mixture is(are) **[2021(MCQ)]**

(A) $Pb(NO_3)_2$ and $Zn(NO_3)_2$ (B) $Pb(NO_3)_2$ and $Bi(NO_3)_3$

(C) $AgNO_3$ and $Bi(NO_3)_3$ (D) $Pb(NO_3)_2$ and $Hg(NO_3)_2$

46. In the reaction given below, the total number of atoms having sp^2 hybridization in the major product P is ______.

[2021(Inte.)]



47. For the following reaction

$$2X + Y \xrightarrow{k} P$$

the rate of reaction is $\frac{d[P]}{dt} = k[X]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution.

At 50 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is(are)

(Use: ln 2 = 0.693) **[2021(MCQ)]**

(A) The rate constant, k, of the reaction is 13.86×10^{-4} s^{-1}.

(B) Half-life of X is 50 s.

(C) At 50 s, $-\frac{d[X]}{dt} = 13.86 \times 10^{-3}$ mol L^{-1} s^{-1}.

(D) At 100 s, $-\frac{d[Y]}{dt} = 3.46 \times 10^{-3}$ mol L^{-1} s^{-1}.

Question Stem for Question Nos. 48 and 49

Question Stem

At 298 K, the limiting molar conductivity of a weak monobasic acid is 4×10^2 S cm^2 mol^{-1}. At 298 K, for an aqueous solution of the acid the degree of dissociation is α and the molar conductivity is $y \times 10^2$ S cm^2 mol^{-1}. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2$ S cm^2 mol^{-1}.

48. The value of α is _______.

49. The value of y is _______. **[2021(Num.)]**

Paragraph For Q. No. 50 & 51

The amount of energy required to break a bond is same as the amount of energy released when the same bond is formed. In gaseous state, the energy required for homolytic cleavage of a bond is called Bond Dissociation Energy (BDE) or Bond Strength. BDE is affected by *s*-character of the bond and the stability of the radicals formed. Shorter bonds are typically stronger bonds. BDEs for some bonds are given below:

$H_3C-H(g) \longrightarrow H_3\dot{C}(g) + \dot{H}(g)\ \Delta H° = 105\ \text{kcal mol}^{-1}$

$Cl-Cl(g) \longrightarrow \dot{Cl}(g) + \dot{Cl}(g)\ \Delta H° = 58\ \text{kcal mol}^{-1}$

$H_3C-Cl(g) \longrightarrow H_3\dot{C}(g) + \dot{Cl}(g)\ \Delta H° = 85\ \text{kcal mol}^{-1}$

$H-Cl(g) \longrightarrow \dot{H}(g) + \dot{Cl}(g)\ \Delta H° = 103\ \text{kcal mol}^{-1}$

50. Correct match of the **C-H** bonds (shown in bold) in Column **J** with their BDE in Column **K** is **[2021(SCQ)]**

	Column J		**Column K**
	Molecule		**BDE (kcal mol^{-1})**
(P)	**H-C**$H(CH_3)_2$	(i)	132
(Q)	**H-C**H_2Ph	(ii)	110
(R)	**H-C**$H=CH_2$	(iii)	95
(S)	**H-C**$\equiv CH$	(iv)	88

(A) P – iii, Q – iv, R – ii, S – i
(B) P – i, Q – ii, R – iii, S – iv
(C) P – iii, Q – ii, R – i, S – iv
(D) P – ii, Q – i, R – iv, S – iii

51. For the following reaction **[2021(SCQ)]**

$$CH_4(g) + Cl_2(g) \xrightarrow{\text{light}} CH_3Cl(g) + HCl(g)$$

the correct statement is

(A) Initiation step is exothermic with $\Delta H° = -58$ kcal mol^{-1}
(B) Propagation step involving $^{\bullet}CH_3$ formation is exothermic with $\Delta H° = -2$ kcal mol^{-1}
(C) Propagation step involving CH_3Cl formation is endothermic with $\Delta H° = +27$ kcal mol^{-1}
(D) The reaction is exothermic with $\Delta H° = -25$ kcal mol^{-1}

52. Consider a helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s^{-1}) of He atom after the photon absorption is_____.
(Assume: Momentum is conserved when photon is absorbed.
Use: Planck constant = 6.6×10^{-34} J s, Avogadro number = 6×10^{23} mol^{-1}, Molar mass of He = 4 g mol^{-1})

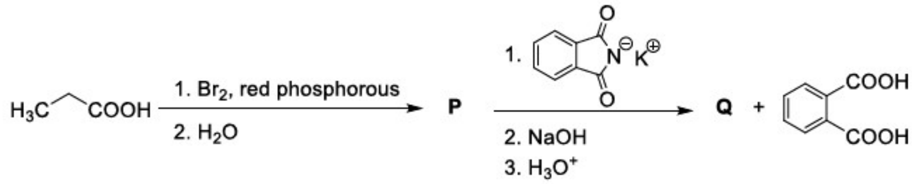
[2021(Int.)]

53. The reduction potential (E^0, in V) of $MnO_4^-(aq)/Mn(s)$ is ______.
[Given: $E^0_{(MnO_4^-(aq)/MnO_2(s))} = 1.68$ V; $E^0_{(MnO_2(s)/Mn^{2+}(aq))} = 1.21$ V; $E^0_{(Mn^{2+}(aq)/Mn(s))} = -1.03$ V]

[2022(Num.)]

54. The treatment of an aqueous solution of 3.74 g of $Cu(NO_3)_2$ with excess KI results in a brown solution along with the formation of a precipitate. Passing H_2S through this brown solution gives another precipitate **X**. The amount of **X** (in g) is ________.
[Given: Atomic mass of H = 1, N = 14, O = 16, S = 32, K = 39, Cu = 63, I = 127] **[2022(Num.)]**

55. Considering the reaction sequence given below, the correct statement(s) is(are) **[2022(MCQ)]**



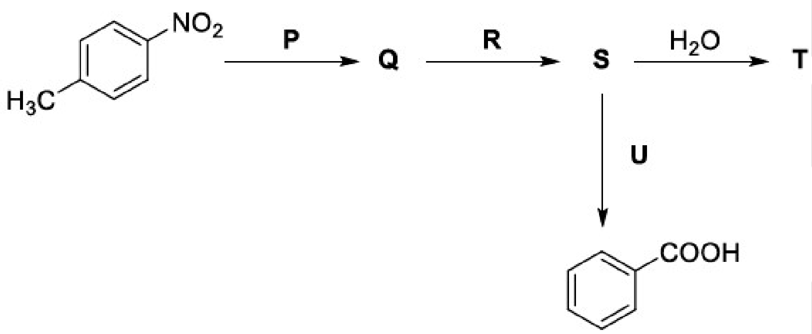
(A) **P** can be reduced to a primary alcohol using $NaBH_4$.

(B) Treating **P** with conc. NH_4OH solution followed by acidification gives Q.

(C) Treating **Q** with a solution of $NaNO_2$ in aq. HCl liberates N_2.

(D) **P** is more acidic than CH_3CH_2COOH.

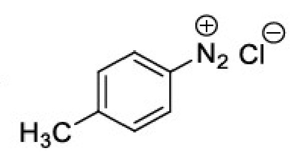
56. Considering the following reaction sequence, **[2022(MCQ)]**

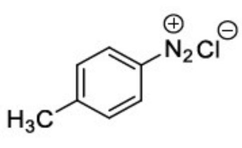
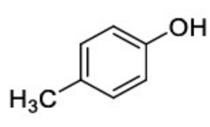


the correct option(s) is(are)

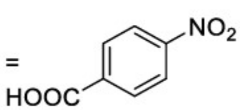
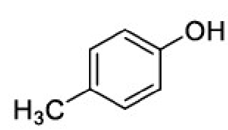
(A) **P** = H_2/Pd, ethanol **R** = $NaNO_2$/HCl **U** = 1. H_3PO_2

2. $KMnO_4$ – KOH, heat

(B) **P** = Sn/HCl **R** = HNO_2 **S** = 

(C) **S** =  **T** =  **U** = 1. CH_3CH_2OH

2. $KMnO_4$ – KOH, heat

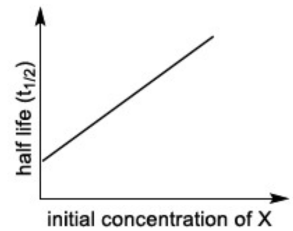
(D) **Q** =  **R** = H_2/Pd, ethanol **T** = 

57. Match the rate expressions in LIST-I for the decomposition of X with the corresponding profiles provided in LIST-II. X_s and k are constants having appropriate units. **[2022(SCQ)]**

LIST-I **LIST-II**

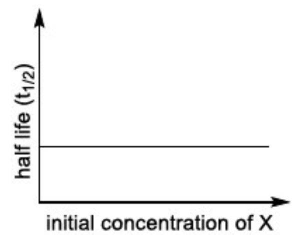
(I) $\text{rate} = \dfrac{k[X]}{X_s + [X]}$

under all possible initial concentrations of X

(P) 

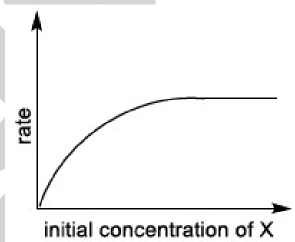
(II) $\text{rate} = \dfrac{k[X]}{X_s + [X]}$

where initial concentrations of X are much less than X_s

(Q) 

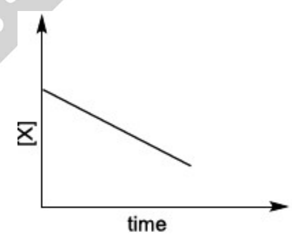
(III) $\text{rate} = \dfrac{k[X]}{X_s + [X]}$

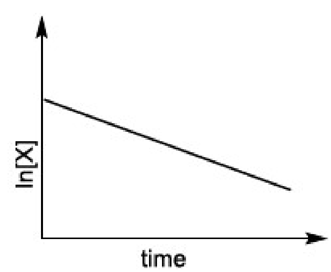
where initial concentrations of X are much higher than X_s

(R) 

(IV) $\text{rate} = \dfrac{k[X]^2}{X_s + [X]}$

where initial concentration of X is much higher than X_s

(S) 

(T) 

(A) I$\rightarrow$ P; II$\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ T

(B) I $\rightarrow$ R; II$\rightarrow$ S; III $\rightarrow$ S; IV $\rightarrow$ T

(C) I$\rightarrow$ P; II$\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ R

(D) I$\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ R

58. LIST-I contains metal species and LIST-II contains their properties. **[2022(SCQ)]**

LIST-I		**LIST-II**	
(I)	$[Cr(CN)_6]^{4-}$	(P)	t_{2g} orbitals contain 4 electrons
(II)	$[RuCl_6]^{2-}$	(Q)	μ(spin-only) = 4.9 BM
(III)	$[Cr(H_2O)_6]^{2+}$	(R)	low spin complex ion
(IV)	$[Fe(H_2O)_6]^{2+}$	(S)	metal ion in 4+ oxidation state
		(T)	d^4 species

[Given: Atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option

(A) I → R, T; II → P, S; III → Q, T; IV → P, Q

(B) I → R, S; II → P, T; III → P, Q; IV → Q, T

(C) I → P, R; II → R, S; III → R, T; IV → P, T

(D) I → Q, T; II → S, T; III → P, T; IV → Q, R

59. Concentration of H_2SO_4 and Na_2SO_4 in a solution is 1 M and 1.8×10^{-2} M, respectively. Molar solubility of $PbSO_4$ in the same solution is $X \times 10^{-Y}$ M (expressed in scientific notation). The value of Y is _________.

[Given: Solubility product of $PbSO_4$ (K_{sp}) = 1.6×10^{-8}. For H_2SO_4, K_{a1} is very large and $Ka_2 = 1.2 \times 10^{-2}$] **[2022(Int.)]**

60. To check the principle of multiple proportions, a series of pure binary compounds (P_mQ_n) were analyzed and their composition is tabulated below. The correct option(s) is(are) **[2022(MCQ)]**

Compound	Weight % of P	Weight % of Q
1	50	50
2	44.4	55.6
3	40	60

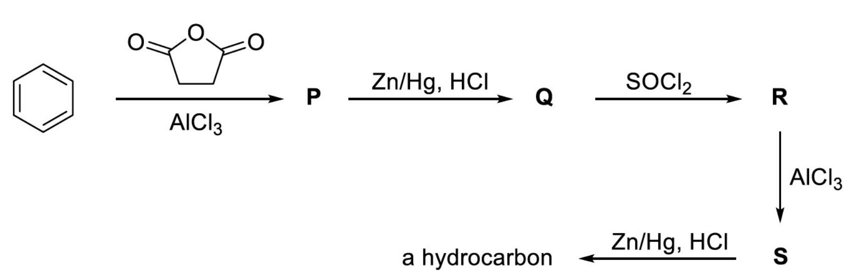
(A) If empirical formula of compound 3 is P_3Q_4, then the empirical formula of compound 2 is P_3Q_5.

(B) If empirical formula of compound 3 is P_3Q_2 and atomic weight of element P is 20, then the atomic weight of Q is 45.

(C) If empirical formula of compound 2 is PQ, then the empirical formula of the compound 1 is P_5Q_4.

(D) If atomic weight of P and Q are 70 and 35, respectively, then the empirical formula of compound 1 is P_2Q.

61. Considering the following reaction sequence, the correct statement(s) is(are) **[2022(MCQ)]**



(A) Compounds P and Q are carboxylic acids.

(B) Compound S decolorizes bromine water.

(C) Compounds P and S react with hydroxylamine to give the corresponding oximes.

(D) Compound R reacts with dialkylcadmium to give the corresponding tertiary alcohol.

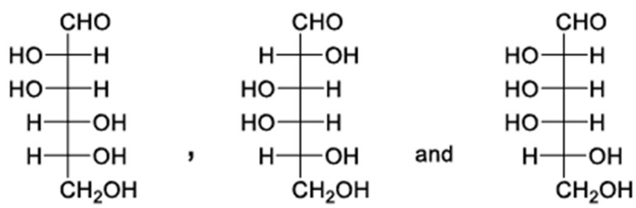
62. Atom X occupies the fcc lattice sites as well as alternate tetrahedral voids of the same lattice. The packing efficiency (in %) of the resultant solid is closest to **[2022(SCQ)]**

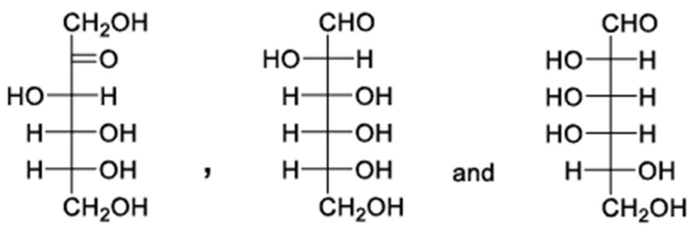
(A) 25 (B) 35

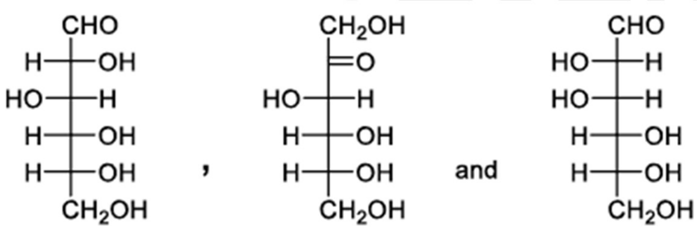
(C) 55 (D) 75

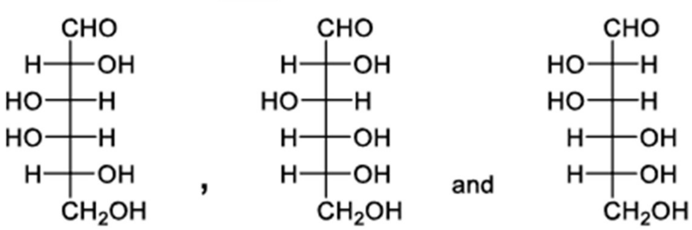
63. Treatment of D-glucose with aqueous NaOH results in a mixture of monosaccharides, which are

[2022(SCQ)]

(A) 

(B) 

(C) 

(D) 

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